Project #5 Report by Tomas Salaz
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Written Problems with Explanations Below

Numerical Computing HW#5

1) Provide a proof for: If points $x_0, x_1, \dots x_n$ are distinct then for arbitrary real values $y_0, y_1, \dots y_n$, there is a unique polynomial, $P_n(x)$ of degree (at most) n s.t. $P_n(x_i) = y_i$, $0 \le i \le n$

- Assume 2 Interpolating Polynomials $P(x)$ and $Q(x)$, both degree n such that for $0 \le i \le n$, $P(x_i) = y_i$ and $Q(x_i) = y_i$

- A 3rd polynomial $R(x) = P(x) - Q(x)$, implies that for all x_i, $0 \le i \le n$ then $R(x_i) = 0$
  ↳ Implies that $R(x)$ has $n+1$ roots, so x_0 to x_n. Since $R(x)$ has degree n, $R(x)$ can have at most n roots, this is per "the Fundamental Thm of Algebra".
- This implies that $R(x)$ is the zero polynomial thus $P(x) = Q(x)$ and is unique for the given set of points

2) Use Lagrange to find Interpolating Polynomial for table below

x	0	2	3	4
y	7	11	28	63

Form $P_3(x) = \sum_{i=0}^{3} l_i(x) f(x_i)$ } from Kincaide page 155

$$l_i(x) = \prod_{\substack{j \ne i \\ j=0}}^{n} \left(\frac{x - x_j}{x_i - x_j} \right)$$

- Without l_i,

$$P_n(x) = 7 \cdot l_0(x) + 11 l_1(x) + 28 l_2(x) + 63 l_3(x)$$

- Finding l_i's

$$l_0 = \left(\frac{x-x_1}{x_0-x_1}\right)\left(\frac{x-x_2}{x_0-x_2}\right)\left(\frac{x-x_3}{x_0-x_3}\right) = \left(\frac{x-2}{0-2}\right)\left(\frac{x-3}{0-3}\right)\left(\frac{x-4}{0-4}\right) = -24(x-2)(x-3)(x-4)$$

$$l_1 = \left(\frac{x-x_0}{x_1-x_0}\right)\left(\frac{x-x_2}{x_1-x_2}\right)\left(\frac{x-x_3}{x_1-x_3}\right) = \left(\frac{x-0}{2-0}\right)\left(\frac{x-3}{2-3}\right)\left(\frac{x-4}{2-4}\right) = 4(x)(x-3)(x-4)$$

$$l_2 = \left(\frac{x-x_0}{x_2-x_0}\right)\left(\frac{x-x_1}{x_2-x_1}\right)\left(\frac{x-x_3}{x_2-x_3}\right) = \left(\frac{x-0}{3-0}\right)\left(\frac{x-2}{3-2}\right)\left(\frac{x-4}{3-4}\right) = -3(x)(x-2)(x-4)$$

$$l_3 = \left(\frac{x-x_0}{x_3-x_0}\right)\left(\frac{x-x_1}{x_3-x_1}\right)\left(\frac{x-x_2}{x_3-x_2}\right) = \left(\frac{x-0}{4-0}\right)\left(\frac{x-2}{4-2}\right)\left(\frac{x-3}{4-3}\right) = 8(x)(x-2)(x-3)$$

$$L_3(x) = -7 \cdot 24(x-2)(x-3)(x-4) + 44(x)(x-3)(x-4) - 3 \cdot 28(x)(x-2)(x-4) + 8 \cdot 63(x)(x-2)(x-3)$$

Numerical Computing HW #5

3) Newtons Interpolation Form

Burden #123 → Divided Difference : $P_n(x) = f[x_0] + \sum_{k=1}^{n} f[x_0, x_1, \dots x_k](x-x_0)\dots(x-x_{k-1})$

	x	$f(x)$	$f[,]$	$f[,,]$	$f[,,,]$
0	0	7	$\frac{11-7}{2-0} = 2$	$\frac{17-2}{3-0} = 5$	$\frac{9-5}{4} = 1$
1	2	11	$\frac{28-11}{3-2} = 17$	$\frac{35-17}{3-1} = 9$	
2	3	28	$\frac{63-28}{4-3} = 35$		
3	4	63			

$a_0 = 7$
$a_1 = 2$
$a_2 = 5$
$a_3 = 1$

$= f(x_0) + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2)$

$P_3(x) = 7 + 2(x-7) + 5(x-7)(x-11) + (x-7)(x-11)(x-63)$

* Polynomial Compairison

$N_3(x) = 7 + 2(x-7) + 5(x-7)(x-11) + (x-7)(x-11)(x-63)$

$L_3(x) = -168(x-2)(x-3)(x-4) + 44(x)(x-3)(x-4) - 84(x)(x-2)(x-4) + 504(x)(x-2)(x-3)$

- The Thm says that for all (x_i, y_i) where $0 \le i \le n$, that the polynomial $P_n(x_i) = y_i$.
- If the polynomials are done correctly then $N_3(x_i) = L_3(x_i) = y_i$ implying that the values are the same at the Nodes. The Theorm never mentions intermediate points, so as long as $N_3(x_i) = L_3(x_i)$ the Uniqueness Thm is true

4) A unique Polynomial, degree ≤ 2, ST $p(0)=0$, $p(1)=1$, $p'(\alpha)=2$ for $\alpha \in [0,1]$ excep the value α, called α_0. Determine α_0 and construct the polynomial for $\alpha \ne \alpha_0$

- $p(x)$ has form $P(x) = ax^2 + bx + c$
- if $p(0) = 0$, $c = 0$ so $\underline{p(x) = ax^2 + bx}$
- if $p(1) = a(1)^2 + b(1) = 1 \Rightarrow \underline{a+b=1}$
- $\alpha_0 = \frac{1}{2}$, Solve for a and b using $2a\alpha + b = 2$ and $a+b=1$,

$a = \frac{1}{2\alpha - 1}$ $b = 1 - \frac{1}{2\alpha - 1}$

$$\boxed{p(x) = \left(\frac{1}{2\alpha-1}\right)x^2 + \left(1 - \frac{1}{2\alpha-1}\right)x}$$

Find $p'(x)$

$P(x) = ax^2 + bx$

$p'(x) = 2ax + b$

- If $p'(\alpha) = 2$

$\underline{2 = 2a\alpha + b}$ solve system for α

$\begin{cases} a+b=1 \Rightarrow b = -a \\ 2a\alpha + b = 2 \end{cases}$

$\hookrightarrow 2a\alpha - a = 2$, $a(2\alpha-1) = 1$,

$2\alpha - 1 \ne 0$

$\boxed{\alpha \ne \frac{1}{2}}$

Numerical Computing HW#5

5) If g interpolates a function @ $x_0, x_1, \dots x_{n-1}$ and
h interpolates @ $x_0, x_1, \dots x_n$

Then $g(x) + \frac{x_0 - x}{x_n - x_0}[g(x) - h(x)]$ interpolates f @ $x_0, x_1, \dots x_n$

- Only need to check $x_0, x_1 \dots x_n$
- $I(x) = g(x) + \frac{x_0 - x}{x_n - x_0}[g(x) - h(x)]$
- Check x_0

$I(x_0) = g(x_0) + \frac{\cancel{x_0 - x_0}^{0}}{x_n - x_0}[g(x) - h(x)] \Rightarrow \underline{I(x_0) = g(x_0) = f(x_0)}$

- Check x_n

$I(x_n) = g(x_n) + \cancel{\frac{x_0 - x_n}{x_n - x_0}}^{-1}[g(x_n) - h(x_n)] \Rightarrow I(x_n) = g(x_n) - g(x_n) + h(x_n)$

so $\underline{I(x_n) = h(x_n) = f(x_n)}$

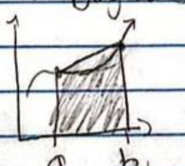
- If $I(x_0) = f(x_0)$ and $I(x_n) = f(x_n)$ the polynomial $I_n(x)$ interpolates the function $f(x)$. Lagrange and Newton Interpolation guarantee that the intermediate points exist

6) Show that the trapezoid rule for Integration results from approximating f by 1st Degree Spline, then using $\int_a^b f(x) \approx \int_a^b S(x)$

Kincaid 203 • Trapezoid Rule : $\int_{x_i}^{x_{i+1}} f(x)dx \approx \frac{1}{2}(x_{i+1} - x_i)[f(x_i) + f(x_{i+1})]$

- 1st Degree spline of $f(x)$ on $[a,b]$
- Spline given by point slope form

Pictoral representation



gives a trapezoid since 1st degree results in a line

$S(x) = f(a) + \frac{f(b) - f(a)}{b-a}(x-a)$

- Taking the integral of $S(x)$: $\int_a^b S(x)$

$= \int_a^b \left(f(a) + \frac{f(b) - f(a)}{b-a}(x-a)\right)dx$

$= \underbrace{\int_a^b f(a)dx}_{} + \underbrace{\int_a^b \frac{f(b) - f(a)}{b-a}(x-a)dx}_{}$

$\int_a^b f(a)dx = f(a)(b-a)$

$\int_a^b \frac{f(b)-f(a)}{b-a}(x-a)dx \quad u = x-a, \; du = dx$

$= \frac{f(b)-f(a)}{b-a}\int_0^{b-a} u\,du$

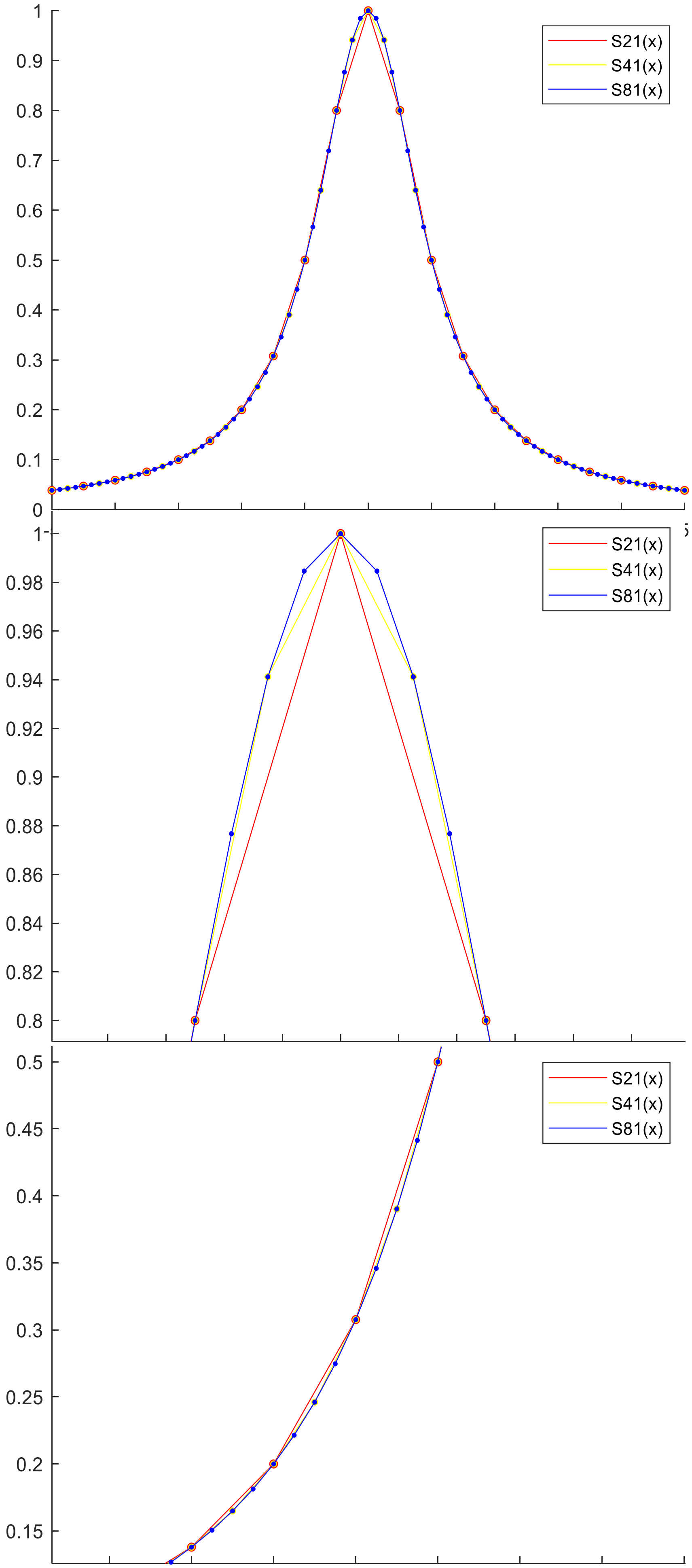
$= \frac{f(b)-f(a)}{b-a} \cdot \frac{(b-a)^2}{2}\Big|_0^{b-a}$

$= \frac{f(b)-f(a)}{b-a}(b-a)$

$= f(a)(b-a) + \frac{f(b) - f(a)}{b-a}(b-a)$

$\boxed{= \left(\frac{f(a) + f(b)}{2}\right)(b-a)} = \int S(x)dx$

Trapezoid rule in a,b terms $\Rightarrow \frac{1}{2}(b-a)(f(b) + f(a))$ ∴ They are the same



Computer Problems

Consider the Runge Function, $f(x) = \frac{1}{1+x^2}$

A) Construct a 1st-Degree Spline using 21,41, and 81 equally spaces nodes on the interval [-5,5]

Summary:
The Top Plot Shows the Graph with all the functions. There is slight variation depending on the knots, especially where the function has dramatic curves. The Bottom 2 graphs show this. These linear interpolations also hinder the differentiability at the knots. It was assumed that more knots would result in a smoother graph, and this is the case. It is also shown that the knots have common values so the S81(x) contains all the S41(x) and the S21(x) knots while S41(x) contains only the S21(x) knots

B)Use 21 ChebyShev Nodes to construct a 1^{st}-Degree Spline and a compare with Normal Nodes

Summary: The ChebyShev Node are used to concentrate the nodes on the Outer ends of the interval to reduce the Error due to oscillation from other interpolations. Since the Spline is only Degree 1 the oscillation doesn't happen so there is little to no error to remove. This can be seen in the bottom 2 plots, the tightly condensed ChebyShev Node are close to the Normal Equally Spaced Nodes at the endpoint. Even near the center of the equations, x=0, the interpolations are similar and it looks like the Equally Spaced Nodes give a more accurate approxamtion.